RISC vs CISC Architecture

1 CISC Architecture

CISC (Complex Instruction Set Computer) processors provide a large and diverse set of instructions. Each instruction can perform multiple operations and may directly access memory.

Key Characteristics

- Large and complex instruction set
- Variable-length instructions
- Instructions can access memory directly
- Fewer instructions required per program
- Complex hardware design
- Difficult to pipeline efficiently

Example of CISC

- Intel 8086
- Intel x86 family (80286, 80386, 80486, Pentium)

Example Instruction

`ADD AX, [BX]`

This single instruction loads data from memory and performs addition.

2 RISC Architecture

RISC (Reduced Instruction Set Computer) processors use a small and simple instruction set. Each instruction performs only one operation and operates on registers only.

Key Characteristics

- Small and simple instruction set
- Fixed-length instructions
- Load-store architecture
- Instructions operate only on registers
- Large number of general-purpose registers
- Faster execution per instruction
- Highly pipeline-friendly

Additional RISC Features

- Separate **LOAD** and **STORE** instructions for memory access
- Hardware-based branch prediction
- Simplified instruction decoding
- Improved performance through pipelining

Examples of RISC Processors

- ARM (Advanced RISC Machine)
- MIPS (Microprocessor without Interlocked Pipeline Stages)
- PowerPC

Example Instructions

```
LDR R1, [R2]
ADD R0, R0, R1
```

3 Comparison Between CISC and RISC

Feature	**CISC**	**RISC**
Instruction Set Size	Large and complex	Small and simple
Instruction Length	Variable	Fixed
Operations per Instruction	Multiple operations	Single operation
Memory Access	Direct memory access allowed	Only via load/store
Registers	Fewer registers	Large number of registers
Execution Speed	Slower per instruction	Faster per instruction
Pipelining	Difficult	Easy and efficient
Hardware Complexity	Complex	Simple
Power Consumption	Higher	Lower
Code Size	Smaller program size	Larger program size
Compiler Complexity	Simple	More complex
Examples	Intel 8086, x86 family	ARM, MIPS, PowerPC

4 Summary

- CISC focuses on reducing the number of instructions per program.
- RISC focuses on faster execution using simple instructions and pipelining.

- Modern processors often combine features of both architectures.